

INDEPENDENT

$$CI_{1-\alpha}(\sigma_Y^2/\sigma_X^2), X \sim N(\mu_X, \sigma_X^2), Y \sim N(\mu_Y, \sigma_Y^2), X \perp Y$$

$X = (X_1, X_2, \dots, X_{m_1})$  IID sample from pop 1

$Y = (Y_1, Y_2, \dots, Y_{m_2})$  IID sample from pop 2

$$\Rightarrow \text{FQ is } W = \frac{(m_1-1)S_X^2/\sigma_X^2 \cdot \frac{1}{m_2-1}}{(m_2-1)S_Y^2/\sigma_Y^2 \cdot \frac{1}{m_1-1}} \sim F_{m_1-1, m_2-1}$$

DEGREES OF FREEDOM

FISHER F DISTRIBUTION  
(RATIO OF TWO "SCALED"  
CHI SQUARES)

USEFUL TO COMPARE THE  
VALIDITY OF MODELS

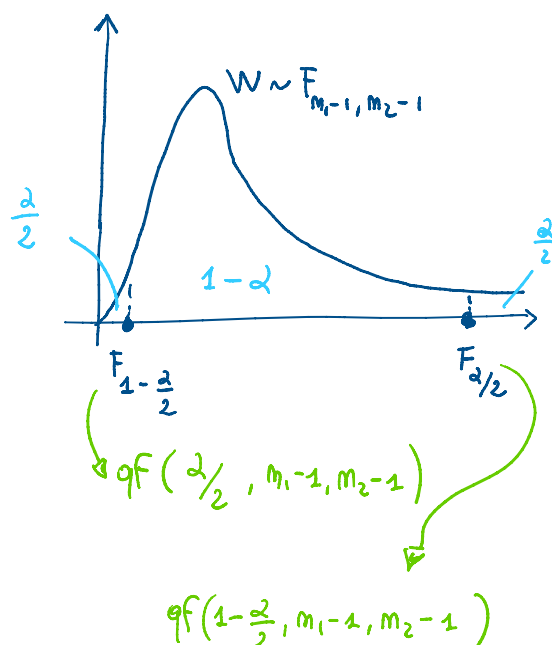
$$= \frac{S_X^2}{S_Y^2} \cdot \frac{\sigma_Y^2}{\sigma_X^2}$$

$$P\left(F_{1-\frac{\alpha}{2}} < W < F_{\alpha/2}\right) = 1-\alpha$$

$$P\left(F_{1-\frac{\alpha}{2}} < \frac{S_X^2}{S_Y^2} \cdot \frac{\sigma_Y^2}{\sigma_X^2} < F_{\alpha/2}\right) = 1-\alpha$$

$$P\left(\frac{S_Y^2}{S_X^2} \cdot F_{1-\frac{\alpha}{2}} < \sigma_Y^2/\sigma_X^2 < \frac{S_Y^2}{S_X^2} \cdot F_{\alpha/2}\right) = 1-\alpha$$

$$\Rightarrow CI_{1-\alpha}(\sigma_Y^2/\sigma_X^2) = \left[ \frac{S_Y^2}{S_X^2} \cdot F_{1-\frac{\alpha}{2}}, \frac{S_Y^2}{S_X^2} \cdot F_{\alpha/2} \right]$$



EX:  $m_1 = 15$ ,  $m_2 = 10$ ,  $S_X^2 = 3.5$ ,  $S_Y^2 = 2.1$ .

find  $CI_{95\%}(\sigma_Y^2/\sigma_X^2)$

$$1-\alpha = 0.95 \Rightarrow \alpha = 0.05 \rightarrow F_{1-\frac{\alpha}{2}} = F_{1-0.025} = F_{0.975} = qf(1-0.025, 15-1, 10-1) = 0.3116$$

$$F_{\frac{\alpha}{2}} = F_{0.025} = qf(1-0.025, 15-1, 10-1) = 3.498$$

$$\Rightarrow C_{(95\%)}(\sigma_1^2/\sigma_2^2) = \left[ \frac{2.1}{3.5} \cdot 0.3116, \frac{2.1}{3.5} \cdot 3.498 \right] = [0.187, 2.249]$$

SINCE  $C \ni 1$ , THEN THERE IS NOT ENOUGH EVIDENCE  
TO SAY THAT THE TWO POP VARIANCES ARE DIFFERENT

• IF  $(1-\alpha) \uparrow \Rightarrow$  NO DIFFERENT CONCLUSION

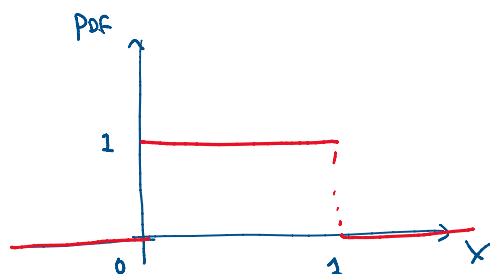
• IF  $(1-\alpha) \downarrow \Rightarrow$  CONCLUSION MAY DIFFER (ACTUAL CALCULATIONS OF THE CI NEEDED)

$$\text{LAPLACE DENSITY: } f(x; \mu, b) = \frac{1}{2b} \exp\left\{-\frac{1}{b}|x-\mu|\right\}, b > 0$$

CLT FOR  $\bar{X}$ , WHEN  $X_i \sim \text{unif}(0,1)$ ,  $i=1, \dots, m$

$$E(X) = 1/2 \Rightarrow E(\bar{X}) = 1/2$$

$$V(X) = 1/12 \Rightarrow V(\bar{X}) = \frac{1}{12m}$$



$\Rightarrow \bar{X} \approx N(1/2, \frac{1}{12m})$  SHOULD WORK  
AS CLT APPROX

BETTER AND BETTER AS WE INCREASE  $m$

IN R: `Xmat`

$N \times m$  MATRIX  
 ↑  
 num of replications      num of RVs on which we compute sample mean

$$\begin{aligned} l(\theta; \mathbf{x}) &= \ln L(\theta; \mathbf{x}) = \ln \prod_{i=1}^m f(x_i; \theta) \\ &= \sum_{i=1}^m \ln f(x_i; \theta) \end{aligned}$$